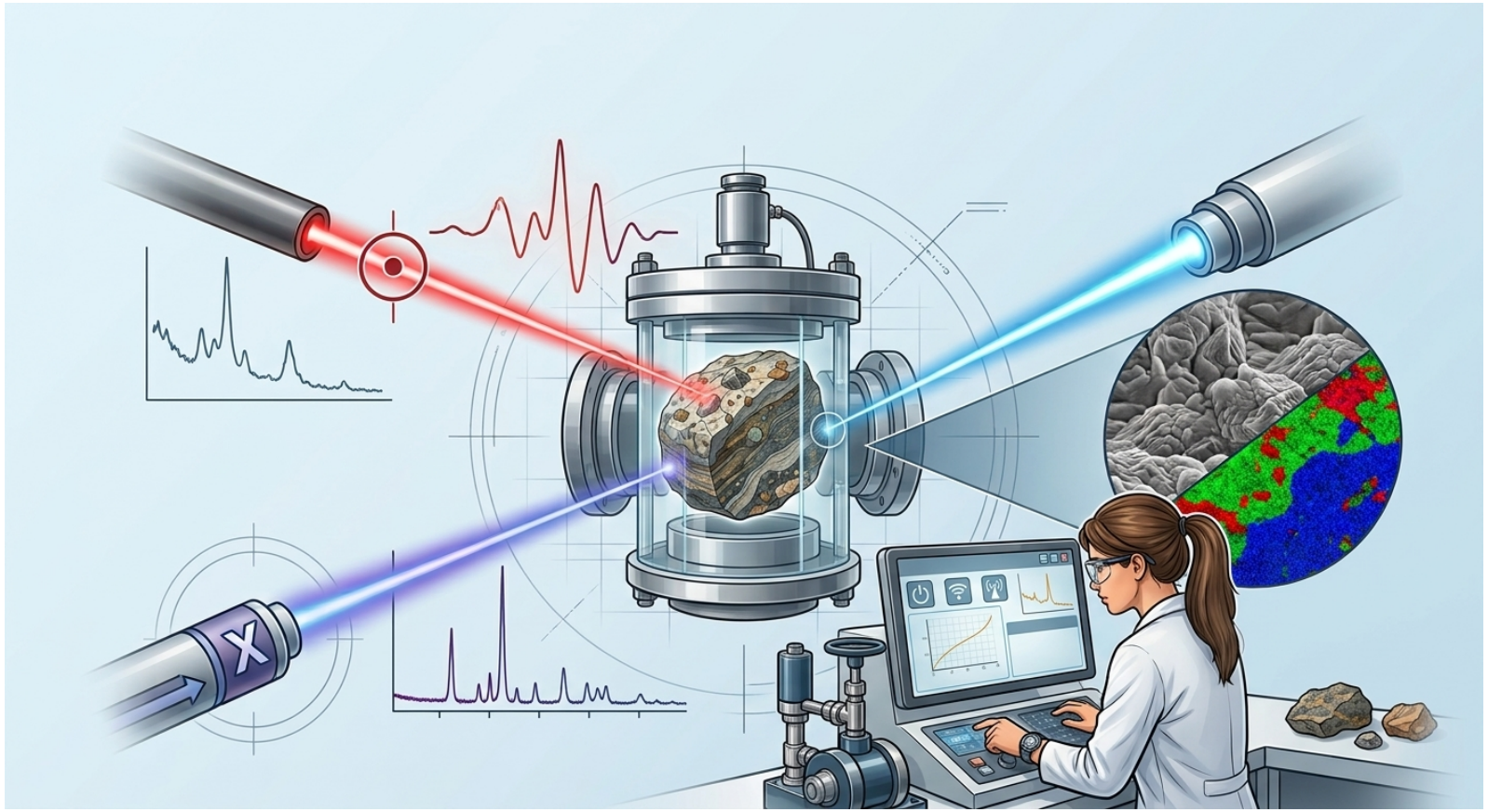
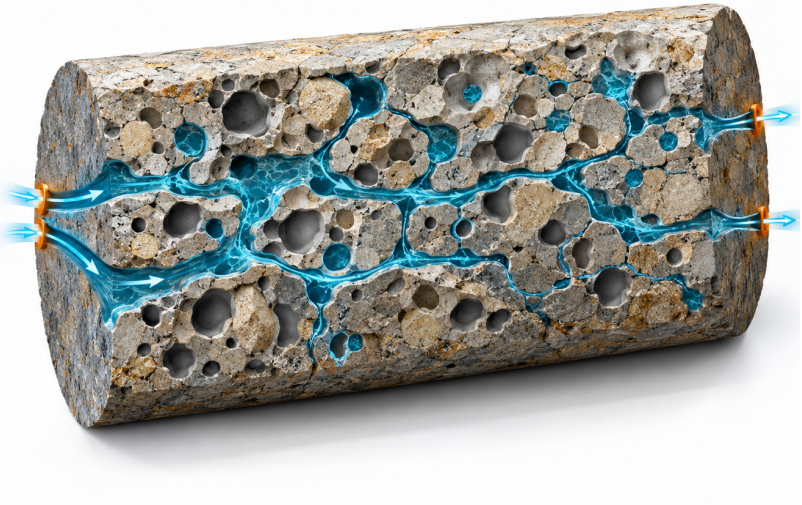
Metodi analitici per lo studio dei geomateriali

Prof.ssa Giorgia Confalonieri
Prof. Marco Scuderi



COURSE INTRODUCTION

Petrophysics



What can a rock sample tell us about a geological event?

Experimental properties, deformation, and fluid–rock interactions

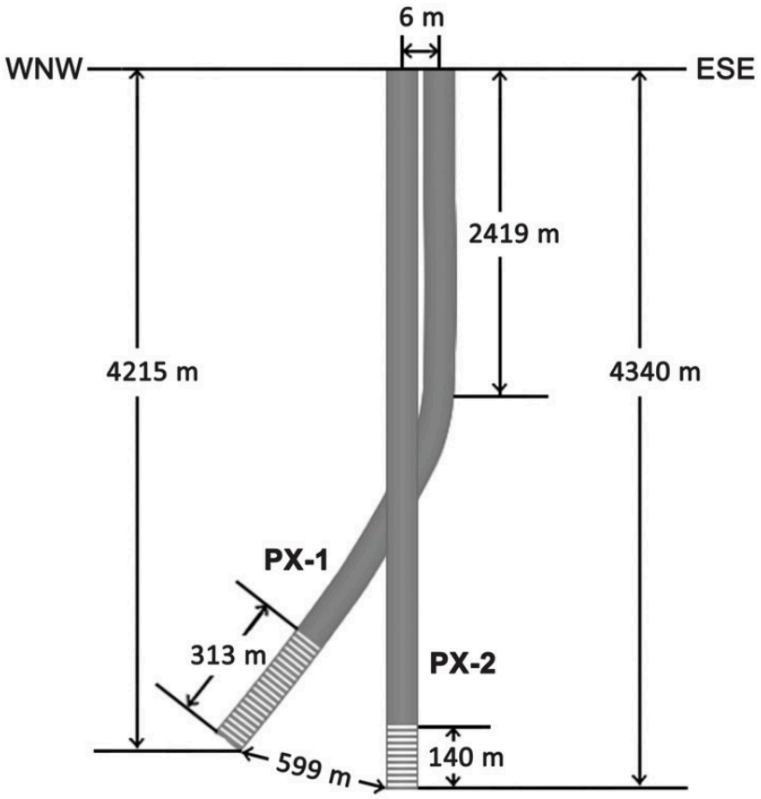
Prof. Marco Scuderi · Sapienza University of Rome

Pohang enhanced geothermal system (EGS) drill site

(a)



(b)

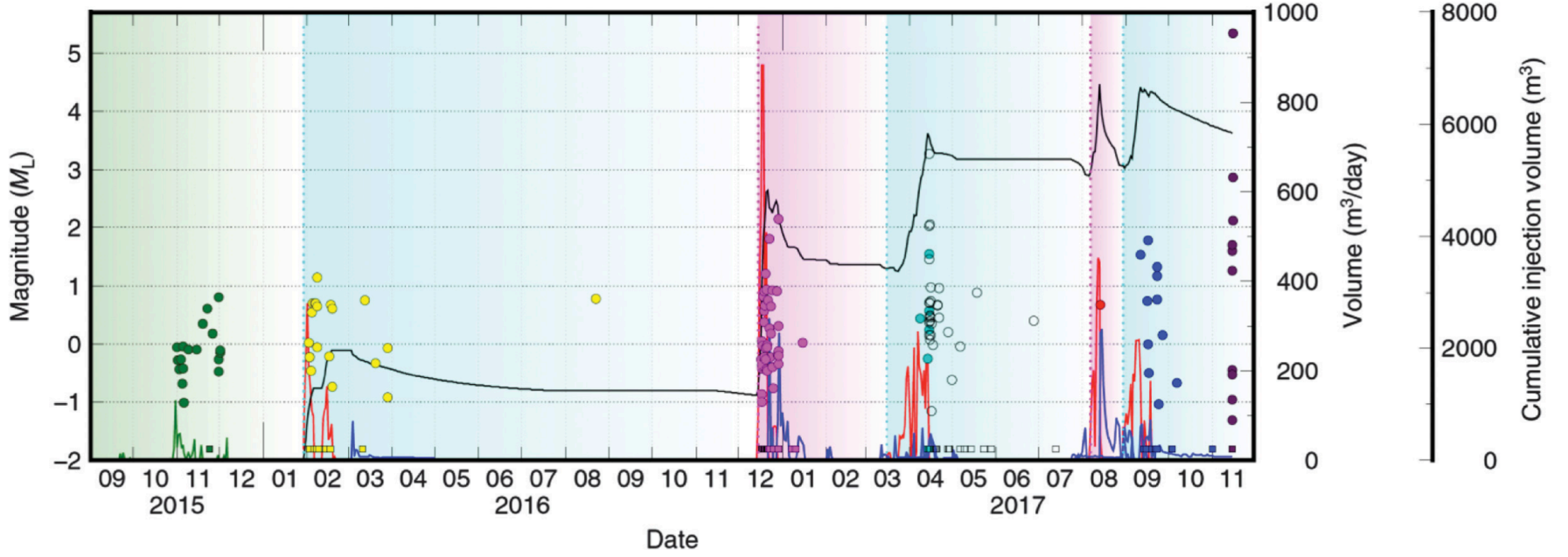


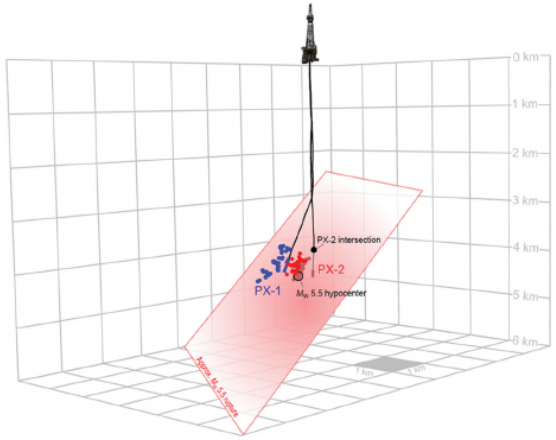
Pohang, South Korea · 15 November 2017

Mw 5.5

An earthquake occurred near an enhanced geothermal project after hydraulic stimulation.

What evidence could link the operations to the earthquake?





What would you investigate first?

Work in pairs. Name two observations you would seek and one alternative explanation.

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2 How could a pressure change reach a fault?

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Work in pairs. Name two observations you would seek and one alternative explanation.

1 What changed in the subsurface?

2 How could a pressure change reach a fault?

3 What evidence would distinguish a trigger from coincidence?

A field interpretation depends on rock measurements

ROCK	FLOW	FAILURE	FIELD
Pores and cracks	Pressure transmission	Strength and damage	Event locations and timing

Every link needs a measurement and an assumption.

Three connected modules

I

Rocks as measurable materials

Structure, stress, strain and ultrasonic waves

II

Damage and brittle failure

Crack growth, acoustic emission and localization

III

Fluids and transport

Permeability, pore pressure and reservoirs

From first measurements to a decision

01–07

MEASURE

Density · mechanics · waves · **UCS test lab** + data analysis

08–13

INTERPRET DAMAGE

Failure · cracks · acoustic emission · **UCS test lab** + data analysis

14–18

FLUIDS

Darcy flow · permeability lab · effective stress · reservoir case